

Data Types and Variables

Python Beginner's Guide

Learning Objectives

After completing this chapter, you will be able to:

- Explain what a variable is.
- Create and update variables in Python.
- Identify common Python data types.
- Use strings, integers, floats, and booleans.
- Check a variable's data type using `type()`.
- Convert values from one type to another.

1. What Is a Variable?

A variable is a name that refers to a value stored by a program. Variables allow us to save information and use it later. In Python, you create a variable by assigning a value with the `=` operator.

Examples:

```
name = "Anmol"  
age = 20  
marks = 85
```

Here, `name`, `age`, and `marks` are variable names. Their values are stored so the program can use them later.

2. Assigning Values to Variables

The assignment operator (`=`) puts a value into a variable. The value on the right side is assigned to the name on the left.

```
city = "Karachi"  
temperature = 32  
is_open = True
```

3. Changing a Variable

A variable can be assigned a new value. The old value is replaced by the new value.

```
score = 50  
print(score)  
  
score = 75  
print(score)
```

4. Rules for Naming Variables

Follow these common rules when naming variables:

- A variable name can contain letters, numbers, and underscores.
- It cannot start with a number.
- Do not use spaces in a variable name.
- Python variable names are case-sensitive.
- Avoid using Python keywords as variable names.

Good examples:

```
student_name = "Ali"  
age2 = 18  
total_marks = 450
```

Poor examples:

```
2age = 18  
student name = "Ali"
```

5. What Is a Data Type?

A data type tells Python what kind of value something is. Different kinds of data can be used for different purposes. Python has several built-in data types.

6. Integer (int)

An integer is a whole number. It can be positive, negative, or zero. Integers do not contain a decimal point.

```
age = 20  
temperature = -5  
students = 30
```

7. Floating-Point Number (float)

A float is a number that contains a decimal value.

```
price = 99.99  
height = 5.8  
percentage = 87.5
```

8. String (str)

A string is text made up of characters. Strings are written inside single quotes or double quotes.

```
name = "Anmol"  
country = 'Pakistan'  
message = "Welcome!"
```

9. Boolean (bool)

A Boolean value represents one of two states: True or False. Booleans are often used when a program needs to make decisions.

```
is_logged_in = True  
has_permission = False
```

10. Checking Data Types with type()

The type() function tells you the data type of a value or variable.

```
age = 20  
name = "Ali"  
height = 5.8  
is_student = True  
  
print(type(age))  
print(type(name))  
print(type(height))  
print(type(is_student))
```

11. Type Conversion

Type conversion means changing a value from one data type to another. Common conversion functions include `int()`, `float()`, `str()`, and `bool()`.

Examples:

```
age_text = "20"
age = int(age_text)

price = "99.50"
price_number = float(price)

number = 100
number_text = str(number)
```

12. Getting Input from the User

The `input()` function allows a program to receive text typed by the user. The value returned by `input()` is a string.

```
name = input("Enter your name: ")
print("Hello", name)
```

If you need a number from the user, convert the input value:

```
age = int(input("Enter your age: "))
print("Your age is", age)
```

13. Multiple Variables

Python allows multiple variables to be assigned in one statement.

```
name, age, city = "Ali", 20, "Karachi"
print(name)
print(age)
print(city)
```

14. Useful Example

The following program stores information about a student and displays it.

```
student_name = "Sara"
age = 18
marks = 92.5
passed = True

print("Name:", student_name)
print("Age:", age)
print("Marks:", marks)
print("Passed:", passed)
```

15. Practice Exercises

Try the following exercises:

- 1 Create variables for your name, age, and city.
- 2 Create an integer variable and a float variable.
- 3 Create a Boolean variable called `is_student`.
- 4 Use `type()` to check the types of your variables.
- 5 Convert the string `'25'` into an integer.
- 6 Ask the user for their age and convert the input to an integer.

16. Mini Project — Student Information

Create a program that asks the user for a name, age, and marks, then displays the information.

Example solution:

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
marks = float(input("Enter your marks: "))

print("Name:", name)
print("Age:", age)
print("Marks:", marks)
```

17. Chapter Summary

Variables are used to store values in Python. Common beginner data types include `int` for whole numbers, `float` for decimal numbers, `str` for text, and `bool` for True or False values. The `type()` function checks a data type, while `int()`, `float()`, `str()`, and `bool()` can be used for type conversion.

Quick Review

- Variables store or refer to values.
- Use `=` to assign a value to a variable.
- `int` represents whole numbers.
- `float` represents decimal numbers.
- `str` represents text.
- `bool` represents True or False.
- `type()` checks the data type.
- `input()` receives user input as text.

End of Chapter